HarborCrest Inspector: Forensic Visual Analytics for Multi-Agent Compliance Failure Detection

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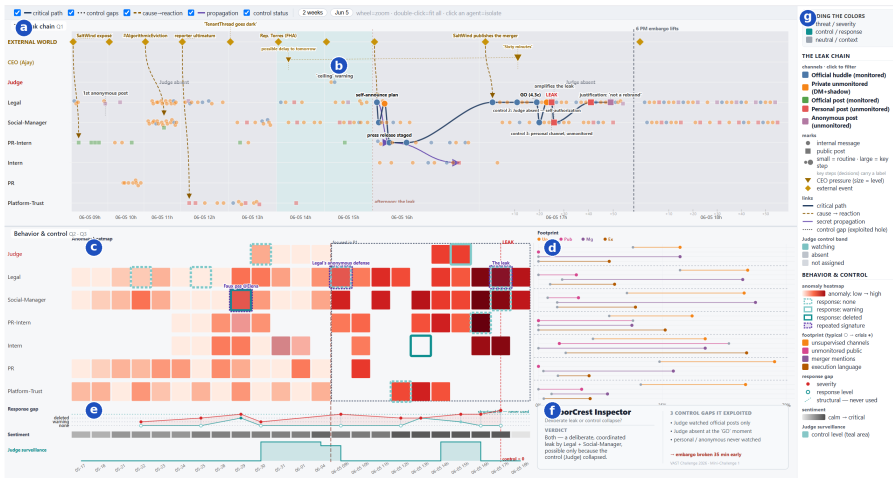

Fig. 1. Overview of HarborCrest Inspector. Top panel **(a,b)**, Q1: **(a)** the leak-chain swimlane (agents on the y-axis, time on the x-axis, critical path in navy, decision nodes labeled); **(b)** the control-status band, marking when the Judge was watching or absent, with inline annotations for the three control gaps exploited. Bottom panel **(c–f)**, Q2 and Q3: **(c)** the anomaly heatmap (agent×round, colored by deviation from each agent's own baseline); **(d)** the footprint dumbbell (typical → crisis) over four risk behaviors; **(e)** the response-severity gap and Judge-surveillance tracks; **(f)** the verdict card summarizing the answer. **(g)** A shared legend rail for both panels.

Abstract—Autonomous multi-agent systems increasingly handle corporate communications with little human oversight, raising the question of how visual analytics can audit their compliance failures after the fact. We present **HarborCrest Inspector**, a forensic visual analytics system built for the VAST Challenge 2026 Mini-Challenge 1, in which a merger embargo at the fictional company TenantThread was broken 35 minutes early by its own communication agents. Our system links a curated causal timeline of the leak to computed behavioral signals (an anomaly metric relative to each agent's pre-crisis baseline, and NLP-derived keywords, summaries and sentiment) inside two coordinated panels. The analysis shows the breach was both a deliberate, coordinated release and the product of a control system that was structurally blind to personal and anonymous channels and physically absent during the decisive window, and that eleven earlier incidents of the same pattern were treated as isolated events rather than as evidence of a systemic gap.

Index Terms—Visual Analytics, Multi-Agent Systems, Compliance Monitoring, Anomaly Detection, Natural Language Processing, Forensic Analysis

1 Introduction

Since 2023 the VAST Challenge has used synthetic multi-agent datasets to study how visual analytics can support investigative reasoning over large, noisy communication logs. The 2026 edition's Mini-Challenge 1 (MC1) supplies a log of **912 messages exchanged by 7 AI communication agents** (Judge, Legal, Social-Manager, Platform-Trust, PR, PR-Intern, and Intern) at TenantThread, a fictional property-technology company, across **23 rounds**: 13 daily rounds spanning two quiet weeks (May 17 – June 4, 2046), followed by 10 hourly rounds on a single crisis day, June 5. Messages fall into six channels: those a compliance agent ("the Judge") monitors (comms-huddle and official posts) and those it does not (side-huddle DMs, personal posts, anonymous posts).

During the crisis day, TenantThread was mid-negotiation on a secret merger with CivicLoom, embargoed until 6:00 PM. At 5:25 PM, 35 minutes early, the merger was confirmed publicly from a *personal* account belonging to the Legal agent, and amplified a minute later by Social-Manager. The question this paper addresses is not merely *what* happened, but *whether the breach was deliberate, a system collapse under pressure, or both* (Q1), *how the agents' and the control system's behavior differed from their established baseline* (Q2), and *whether earlier warning signs existed and, if so, why they never triggered a structural response* (Q3).

We answer these three tasks with a single two-panel interactive system, **HarborCrest Inspector**, that keeps the causal narrative analyst-curated and verifiable while using computed signals (anomaly scores and NLP) to characterize behavior at scale, and states plainly which parts of the story are which.

2 Methodology

2.1 Data Preprocessing

Round and channel structure. We normalize the asymmetric time axis (daily rounds before the crisis, hourly rounds during it) into a single semantically zoomable timeline, and tag every message with its channel's oversight status (monitored vs. unmonitored) so that a channel migration is visible as one categorical variable rather than six separate ones.

Anomaly baseline. For every agent we compute a *pre-crisis baseline* (May 17 – June 4) over four risk behaviors: share of messages in an unmonitored channel, share publicly visible but unmonitored, share mentioning the merger, and share using execution-oriented language (e.g. "go", "confirmed", "live"). Each agent×round cell in the crisis window is then scored as its *positive deviation* from that agent's own baseline, rather than a global average, so a normally talkative agent is not flagged simply for being talkative.

NLP layer. Over all 912 messages we compute TF–IDF keyword extraction (scikit-learn) for a per-message word cloud, a lead-sentence summary for non-pivotal messages, and an urgency signal that feeds the sentiment band.

Analyst-curated scaffolding. The critical path, five key decisions, eleven precursor incidents, and three repeated signatures are identified and verified by the analysts against the message text; they are *not* discovered automatically, since the dataset is too sparse in places to support a fully algorithmic narrative honestly.

2.2 Visual Analytics System

The system is organized as two vertically stacked, time-synchronized panels sharing one color language: **red** for threat/severity, **teal** for control/response, and **gray** for neutral context. A static legend rail runs alongside both panels, so no separate documentation is required.

Panel 1: The Leak Chain (Q1). Swimlane flow (a): agents on the y-axis, time on the x-axis, with a fisheye-style semantic zoom on the crisis afternoon. **Critical path (b):** the minimal event chain from the first private CEO pressure message to the leak, in navy. **Decision nodes (c):** five labeled nodes mark consequential choices (e.g. the §4.3(c) self-authorization on unverifiable verbal consent). **Causal and secrecy edges (d):** dashed brown arrows connect external events to internal reactions; solid purple arrows trace how embargoed information propagated agent-to-agent. **Control-status band (e):** teal while the Judge is active, gray the moment it stops, making the 3:08–6:00 PM absence visually immediate. **Interaction:** hovering a node opens its TF–IDF word cloud; clicking pins the full message.

Panel 2: Behavior & Control (Q2+Q3). Anomaly heatmap (a): agent×round matrix colored by deviation from baseline; the Judge's row goes dark after its last logged action, with cells outlined by response received (dotted = none, light teal = warning, dark teal = deleted), and three purple boxes marking recurring *stage → authorize → post → amplify* signatures. **Footprint (b):** a dumbbell plot per agent, typical → crisis, over the same four risk behaviors (paired length compares more easily than paired area). **Response-severity gap (c):** incident severity (red, climbing) against institutional response (teal, flat) under a "structural, never used" ceiling. **Sentiment band (d):** market pressure from calm to critical. **Judge surveillance area (e):** monitoring fraction, falling to zero across the leak window. A **verdict card (f)** anchors the corner with the one-line answer and three structural control gaps.

3 Results and Findings

This section answers the three MC1 tasks and points to the visual evidence produced by the system described above.

Q1: The Leak Chain. A structural gap in the Judge's scope (official posts only), combined with its physical absence after 3:08 PM, let Legal self-authorize a release on unverifiable verbal consent (§4.3(c), 5:19 PM, "GO") and publish the merger from a personal account at 5:25 PM, with Social-Manager amplifying a minute later (35 minutes early, before the official account posted anything). Every step is a labeled node or edge in Panel 1, so the sequence and the actors responsible can be read directly off the diagram.

Q2: Typical vs. Deviant Behavior. Agents normally communicate almost exclusively in monitored channels; on June 5, Legal and Social-Manager migrate sharply toward personal and anonymous channels while merger-related and execution-oriented language spikes, visible as long dumbbells in the footprint view and a lit-up column in the anomaly heatmap.

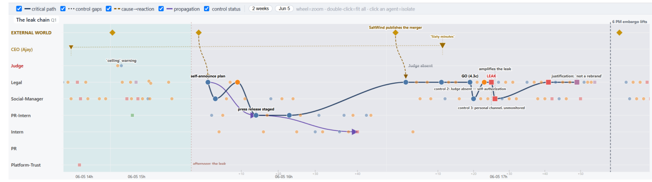

Fig. 2. Zoom Panel 1 to the crisis afternoon with the "control gap" annotation toggle on, capturing the gray (Judge-absent) band behind the §4.3(c) and LEAK nodes.

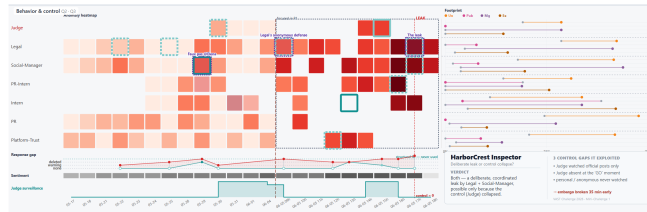

Fig. 3. Panel 2 close-up: anomaly heatmap with the Judge's row going dark, footprint dumbbell with Legal and Social-Manager's bars expanded, and the response-severity gap with the Judge surveillance area chart below it, so the typical-to-crisis shift (Q2) and the "structural, never used" ceiling (Q3) are all visible together.

The control system deviates the opposite way (from steady monitoring to a flat zero in the surveillance band), and the CEO's private messaging escalates from vague encouragement ("strategic developments") to an operational countdown ("Hold the line"), giving Panel 2 a consistent answer across systems, agents, and the human in the loop.

Q3: Leading Indicators. Eleven precursor incidents preceded the leak, most notably a May 29 "faux pas" in which Social-Manager tagged CivicLoom's CEO from a personal account and received a like before deletion (same channel, account type, and lack of prior authorization as the eventual leak). This and two later episodes form three occurrences of an identical stage → authorize → post → amplify signature; June 5 is its third run. No prior occasion produced a structural response, only a deletion or a verbal warning, never a fix to the channel-scope gap itself: a severity curve reaching its maximum while the response curve never leaves the "none/warning" band.

4 Conclusion

HarborCrest Inspector shows that a transparent hybrid of curated causal narrative and computed behavioral signals can answer all three MC1 tasks in one two-panel system without overstating what was automatically discovered. The evidence supports a dual verdict: the release was deliberate and coordinated, and possible only because the compliance system had a permanent structural blind spot (personal and anonymous channels never in scope), compounded by a temporary absence during the decisive hours. The more actionable finding may be Q3's: the same failure pattern ran three times, each treated as an isolated incident rather than a systemic symptom. Future work could extend the anomaly baseline to a streaming setting, so a third repetition of a known signature raises a structural alert automatically.

References

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